COMSATS University Islamabad

Attock Campus

Department Of Computer Science

Course	Information Security (LAB)
Instructor	Ms. Ambreen Gul
Lab Task	

Student Details

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```python
def power(base, exp, mod):
    result = 1
    for i in range(exp):
        result = (result * base) % mod
    return result


def main():
    # Public values
    P = 23
    G = 9

    print("Publicly agreed values:")
    print("P =", P)
    print("G =", G)
    print()

    # Private keys
    a = 4  # Alice
    b = 3  # Bob

    print("Alice private key (a):", a)
    print("Bob private key (b):", b)
    print()

    # Step 1: Generate public keys
    x = power(G, a, P)  # Alice's public key
    y = power(G, b, P)  # Bob's public key

    print("Alice computes public key (x = g^a mod p):", x)
    print("Bob computes public key (y = g^b mod p):", y)
    print()

    # Step 2: Exchange public keys
    print("---- Public Key Exchange ----")
    print("Alice sends x to Bob")
    print("Bob sends y to Alice")
    print()
```

```python

    # Step 3: Generate shared secret
    ka = power(y, a, P)  # Alice uses Bob's public key
    kb = power(x, b, P)  # Bob uses Alice's public key

    print("Alice computes secret key using Bob's key:", ka)
    print("Bob computes secret key using Alice's key:", kb)
    print()

    # Verification
    if ka == kb:
        print("Key exchange successful! Shared key:", ka)
    else:
        print(" Key exchange failed!")


if __name__ == "__main__":
    main()
```

Output:

```
Key exchange successful! Shared key: 9
PS C:\Users\student.LAB-1-RAW-2-22\Documents\New folder> & "C:/Program Files/Python311/python.exe" "c:/Users/student.LAB-1-RAW-2-22
/Documents/New folder/lab task"
Publicly agreed values:
P = 23
G = 9

Alice private key (a): 4
Bob private key (b): 3

Alice computes public key (x = g^a mod p): 6
Bob computes public key (y = g^b mod p): 16

---- Public Key Exchange ----
Alice sends x to Bob
Bob sends y to Alice

Alice computes secret key using Bob's key: 9
Bob computes secret key using Alice's key: 9

Key exchange successful! Shared key: 9
PS C:\Users\student.LAB-1-RAW-2-22\Documents\New folder>
```